

Hands On Unit_3 Descriptive Statistics

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Course Code:DSA0405

Course Name:Fundamentals Of Data Science

Data Science Lab Programs

1. Employee Dataset Summary

Aim:

To generate a complete summary of an employee dataset showing rows, columns, data types, missing values and statistical summary.

Algorithm:

1. Start
2. Import pandas
3. Load employee dataset
4. Display rows and columns
5. Display data types
6. Display missing values
7. Display statistical summary
8. Stop

Code:

```
import pandas as pd

df = pd.read_csv("employee.csv")

print("Number of Rows :", df.shape[0])

print("Number of Columns :", df.shape[1])

print("\nData Types")

print(df.dtypes)

print("\nMissing Values")

print(df.isnull().sum())

print("\nStatistical Summary")

print(df.describe(include='all'))
```

Output:

```
p99.py> ...
1 import pandas as pd
2
3 # Load dataset
4 df = pd.read_csv("employee.csv")
5
6 print("Number of Rows :", df.shape[0])
7 print("Number of Columns :", df.shape[1])
8
9 print("\nData Types")
10 print(df.dtypes)
11
12 print("\nMissing Values")
13 print(df.isnull().sum())
14
15 print("\nStatistical Summary")
16 print(df.describe(include='all'))
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

saicharan@sais-MacBook-Air vscode_practice % python3 p99.py

	Employee_Name	EmpID	MarriedID	MaritalStatusID	...	SpecialProjectsCount	LastPerformanceReview_Date	DaysLateLast30	Absences
count	311	311.000000	311.000000	311.000000	...	311.000000	311	311.000000	311.000000
unique	311	NaN	NaN	NaN	...	NaN	137	NaN	NaN
top	Adinolfi, Wilson	NaN	NaN	NaN	...	NaN	1/14/2019	NaN	NaN
freq	1	NaN	NaN	NaN	...	NaN	10	NaN	NaN
mean	NaN	10156.000000	0.392714	0.810209	...	1.218950	NaN	0.414791	10.237942
std	NaN	89.922189	0.490423	0.943239	...	2.149421	NaN	1.294519	5.852596
min	NaN	10001.000000	0.000000	0.000000	...	0.000000	NaN	0.000000	1.000000
25%	NaN	10070.500000	0.000000	0.000000	...	0.000000	NaN	0.000000	5.000000
50%	NaN	10156.000000	0.000000	1.000000	...	0.000000	NaN	0.000000	10.000000
75%	NaN	10233.500000	1.000000	1.000000	...	0.000000	NaN	0.000000	15.000000
max	NaN	10311.000000	1.000000	4.000000	...	8.000000	NaN	6.000000	20.000000

[11 rows x 10 columns]

Result:

The employee dataset was successfully summarized.

2. Frequency Distribution of Grades

Aim:

To calculate the frequency distribution of students' grades and display a bar chart.

Algorithm:

1. Start
2. Import libraries
3. Load student dataset
4. Create grade frequencies
5. Plot bar chart
6. Stop

Code:

```
import pandas as pd

import matplotlib.pyplot as plt

data = {
    'Grade': ['A','B','A','C','B','A','D','B','C','A']
}

df = pd.DataFrame(data)

frequency = df['Grade'].value_counts()

print(frequency)

frequency.plot(kind='bar')

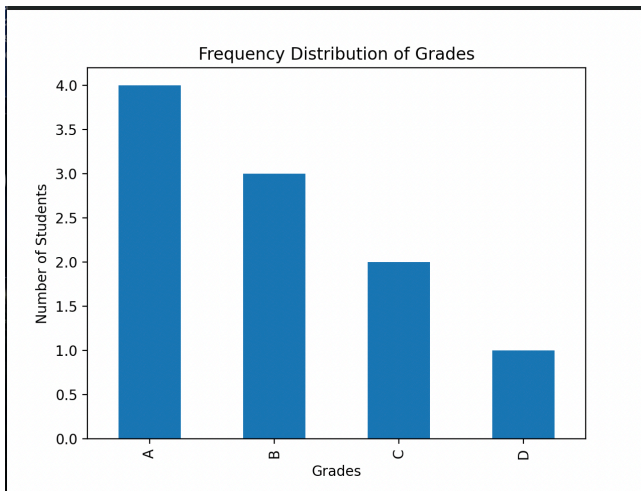
plt.title("Frequency Distribution of Grades")

plt.xlabel("Grades")

plt.ylabel("Number of Students")

plt.show()
```

Output:



Result:

The grade distribution was successfully visualized.

3. Histogram of House Prices

Aim:

To plot a histogram of house prices and observe the distribution.

Algorithm:

1. Start
2. Import libraries
3. Load house dataset
4. Plot histogram
5. Observe distribution
6. Stop

Code:

```
import pandas as pd

import matplotlib.pyplot as plt

data = {

    'Price':[200000,250000,300000,350000,400000,450000,

             500000,550000,600000,900000]

}

df = pd.DataFrame(data)

plt.hist(df['Price'], bins=5)

plt.title("House Price Distribution")

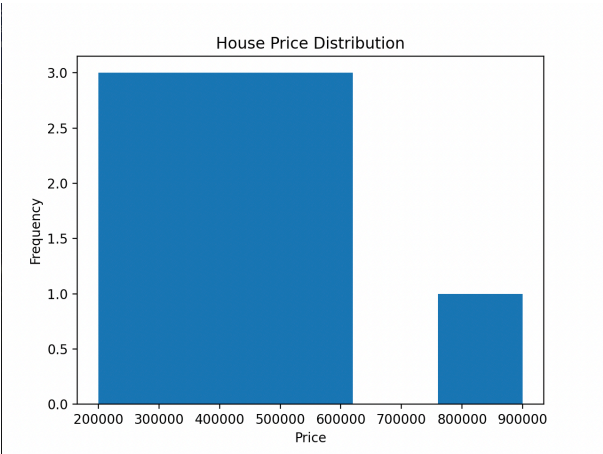
plt.xlabel("Price")

plt.ylabel("Frequency")

plt.show()

print("Observe the histogram to determine whether the data is normal or skewed.")
```

Output:



Result:

The price distribution was visualized.

4. Box Plot of Salary Data

Aim:

To create a box plot of salary data and identify outliers.

Algorithm:

1. Start
2. Import libraries
3. Load salary dataset
4. Plot box plot
5. Identify outliers
6. Stop

Code:

```
import pandas as pd

import matplotlib.pyplot as plt

data = {

    'Salary':[25000,28000,30000,32000,35000,37000,

              40000,42000,45000,100000]

}

df = pd.DataFrame(data)

plt.boxplot(df['Salary'])

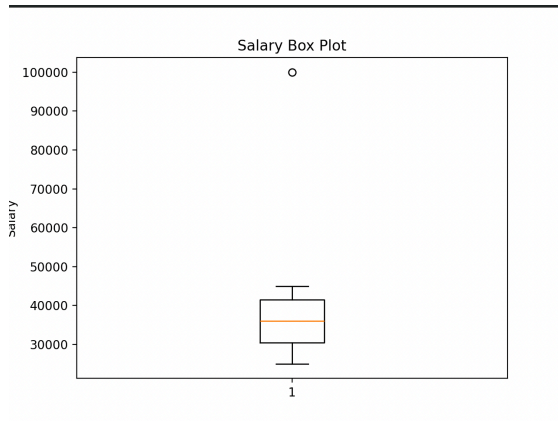
plt.title("Salary Box Plot")

plt.ylabel("Salary")

plt.show()

print("The point outside the whiskers is an outlier.")
```


Output:



Result:

The salary outliers were identified.

5. Detect Outliers using IQR

Aim:

To detect outliers in students' marks using the IQR method.

Algorithm:

1. Start
2. Import pandas
3. Load marks dataset
4. Calculate Q1,Q3,IQR
5. Detect outliers
6. Display outliers
7. Stop

Code:

```
import pandas as pd

data = {
    'Marks':[45,50,52,55,60,62,65,70,75,150]
}

df = pd.DataFrame(data)

Q1 = df['Marks'].quantile(0.25)
Q3 = df['Marks'].quantile(0.75)
IQR = Q3 - Q1
lower = Q1 - 1.5*IQR
upper = Q3 + 1.5*IQR
outliers = df[(df['Marks'] < lower) | (df['Marks'] > upper)]
print("Outliers")
print(outliers)
```

Output:

```
p103.py > ...
1  import pandas as pd
2
3  data = {
4      'Marks': [45,50,52,55,60,62,65,70,75,150]
5  }
6
7  df = pd.DataFrame(data)
8
9  Q1 = df['Marks'].quantile(0.25)
10 Q3 = df['Marks'].quantile(0.75)
11
12 IQR = Q3 - Q1
13
14 lower = Q1 - 1.5*IQR
15 upper = Q3 + 1.5*IQR
16
17 outliers = df[(df['Marks'] < lower) | (df['Marks'] > upper)]
18
19 print("Outliers")
20 print(outliers)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
saicharan@sais-MacBook-Air vscode_practice % python3 p103.py
Outliers
Marks
9      150
```

Result:

Outliers were successfully detected using IQR.

6. Remove Outliers using IQR

Aim:

To remove outliers from a sales dataset using the IQR method.

Algorithm:

1. Start
2. Import pandas
3. Load sales dataset
4. Calculate IQR
5. Remove outliers
6. Display cleaned dataset
7. Stop

Code:

```
import pandas as pd

data = {
    'Sales':[100,120,130,150,170,180,200,210,220,600]
}

df = pd.DataFrame(data)

Q1 = df['Sales'].quantile(0.25)
Q3 = df['Sales'].quantile(0.75)
IQR = Q3 - Q1
lower = Q1 - 1.5*IQR
upper = Q3 + 1.5*IQR
clean_df = df[(df['Sales'] >= lower) & (df['Sales'] <= upper)]

print("Original Dataset")
print(df)
print("\nCleaned Dataset")
print(clean_df)
```

Output:

```
p104.py > ...
1  import pandas as pd
2
3  data = {
4      'Sales': [100, 120, 130, 150, 170, 180, 200, 210, 220, 600]
5  }
6
7  df = pd.DataFrame(data)
8
9  Q1 = df['Sales'].quantile(0.25)
10 Q3 = df['Sales'].quantile(0.75)
11
12 IQR = Q3 - Q1
13
14 lower = Q1 - 1.5*IQR
15 upper = Q3 + 1.5*IQR
16
17 clean_df = df[(df['Sales'] >= lower) & (df['Sales'] <= upper)]
18
19 print("Original Dataset")
20 print(df)
21
22 print("\nCleaned Dataset")
23 print(clean_df)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

saicharan@sais-MacBook-Air vscode_practice % python3 p104.py

```
8  220
9  600

Cleaned Dataset
   Sales
0    100
1    120
2    130
3    150
4    170
5    180
6    200
7    210
8    220
```

Result:

Outliers were removed successfully.

7. Histograms and Box Plots Before and After

Aim:

To compare data before and after removing outliers.

Algorithm:

1. Start
2. Import libraries
3. Load dataset
4. Plot before graphs
5. Remove outliers
6. Plot after graphs
7. Stop

Code:

```
import pandas as pd

import matplotlib.pyplot as plt

data = {
    'Value': [20, 22, 25, 27, 30, 32, 35, 37, 40, 100]
}

df = pd.DataFrame(data)

Q1 = df['Value'].quantile(0.25)
Q3 = df['Value'].quantile(0.75)
IQR = Q3 - Q1
lower = Q1 - 1.5*IQR
upper = Q3 + 1.5*IQR
clean_df = df[(df['Value'] >= lower) & (df['Value'] <= upper)]

plt.hist(df['Value'])

plt.title("Histogram Before Removing Outliers")

plt.show()

plt.hist(clean_df['Value'])

plt.title("Histogram After Removing Outliers")

plt.show()
```

```
plt.boxplot(df['Value'])

plt.title("Box Plot Before Removing Outliers")

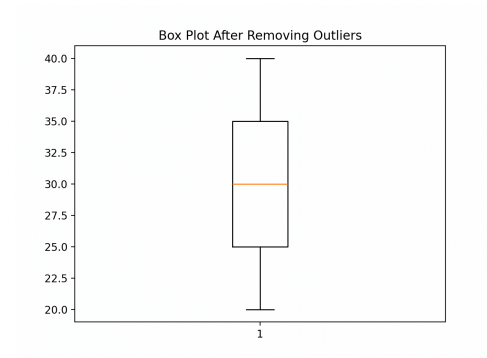
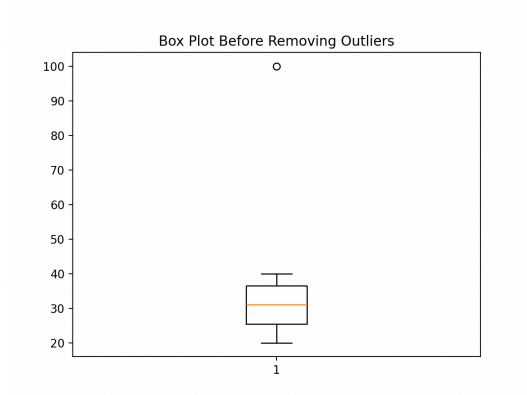
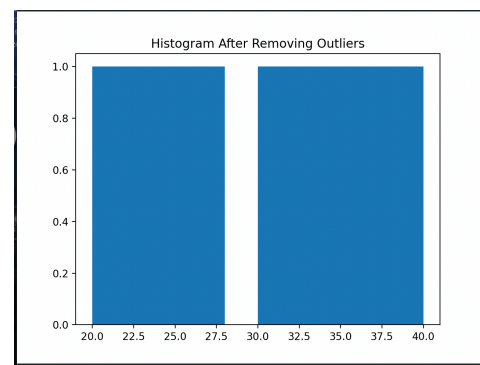
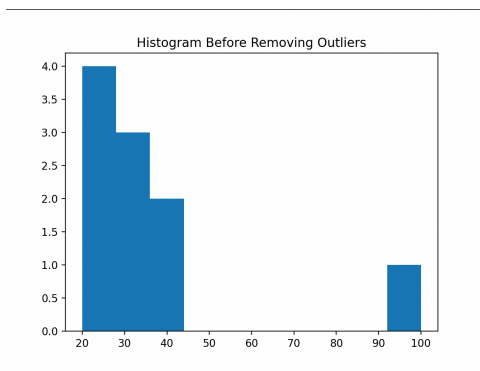
plt.show()

plt.boxplot(clean_df['Value'])

plt.title("Box Plot After Removing Outliers")

plt.show()
```

Output:



Result:
The effect of outlier removal was visualized.

8. Exploratory Data Analysis (EDA)

Aim:

To perform Exploratory Data Analysis on the Iris/Titanic dataset.

Algorithm:

1. Start
2. Load dataset
3. Display info
4. Find missing values
5. Compute statistics
6. Plot graphs
7. Detect/remove outliers
8. Save cleaned dataset
9. Stop

Code:

```
import pandas as pd

import matplotlib.pyplot as plt

df = pd.read_csv("Titanic-Dataset.csv")

print("===== DATASET INFORMATION =====")

print(df.info())

# Step 3: Display first 5 rows

print("\n===== FIRST 5 ROWS =====")

print(df.head())

print("\n===== MISSING VALUES =====")

print(df.isnull().sum())

print("\n===== DESCRIPTIVE STATISTICS =====")

print(df.describe())

numeric_columns = df.select_dtypes(include=['int64', 'float64']).columns

for column in numeric_columns:

    plt.figure(figsize=(6,4))

    plt.hist(df[column].dropna(), bins=15)

    plt.title("Histogram of " + column)
```



```

plt.xlabel(column)

plt.ylabel("Frequency")

plt.show()

for column in numeric_columns:

    plt.figure(figsize=(5,4))

    plt.boxplot(df[column].dropna())

    plt.title("Box Plot of " + column)

    plt.ylabel(column)

    plt.show()

clean_df = df.copy()

for column in numeric_columns:

    Q1 = clean_df[column].quantile(0.25)

    Q3 = clean_df[column].quantile(0.75)

    IQR = Q3 - Q1

    lower_limit = Q1 - 1.5 * IQR

    upper_limit = Q3 + 1.5 * IQR

    outliers = clean_df[

        (clean_df[column] < lower_limit) |

        (clean_df[column] > upper_limit)

    ]

    print("\nOutliers in", column, ":", len(outliers))

for column in numeric_columns:

    Q1 = clean_df[column].quantile(0.25)

    Q3 = clean_df[column].quantile(0.75)

    IQR = Q3 - Q1

    lower_limit = Q1 - 1.5 * IQR

    upper_limit = Q3 + 1.5 * IQR

```

```

clean_df = clean_df[

(clean_df[column] >= lower_limit) &

(clean_df[column] <= upper_limit)

]

print("\nOriginal Dataset Shape :", df.shape)

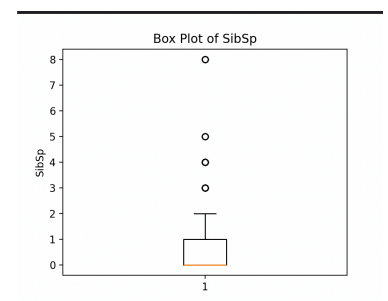
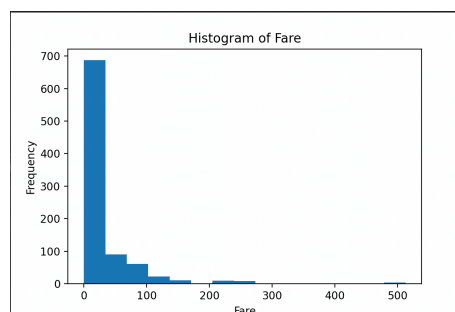
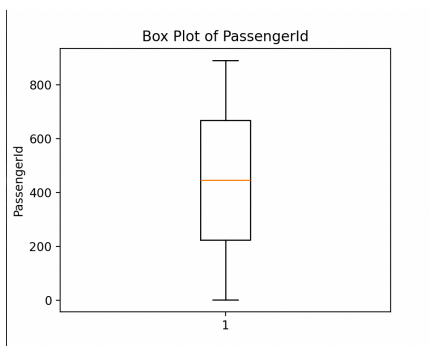
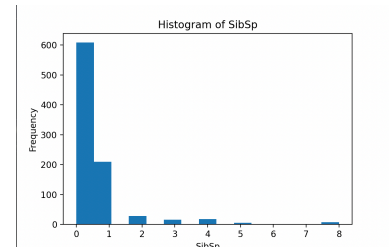
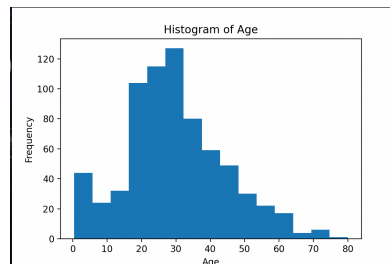
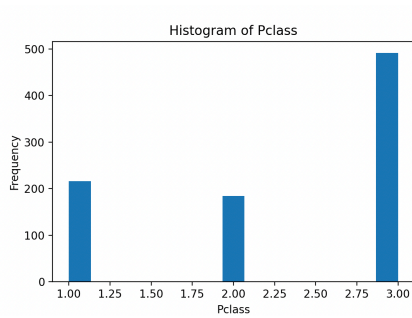
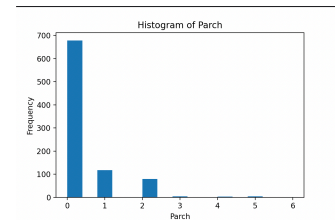
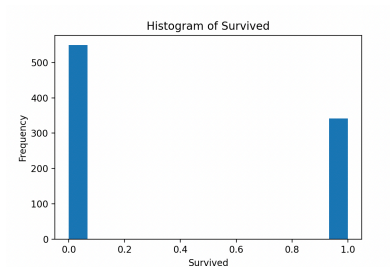
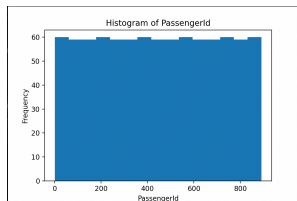
print("Cleaned Dataset Shape :", clean_df.shape)

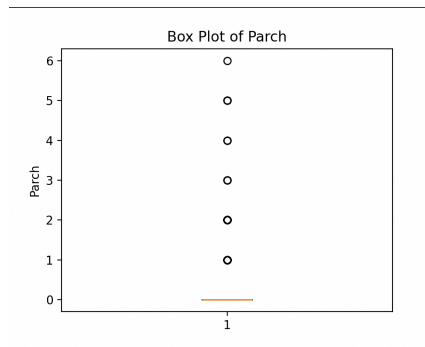
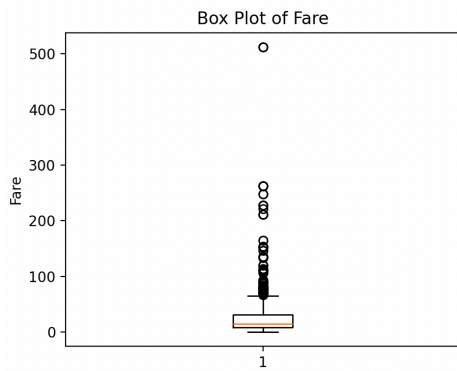
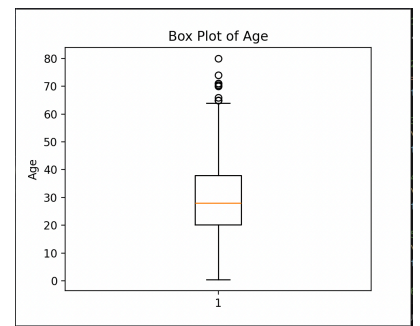
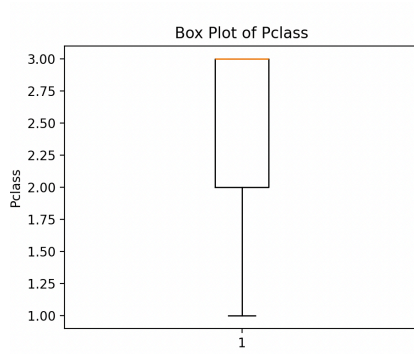
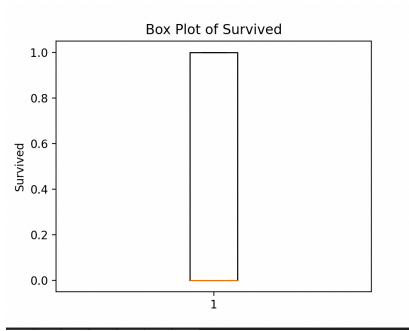
clean_df.to_csv("Titanic_Cleaned.csv", index=False)

print("\nCleaned dataset saved successfully as Titanic_Cleaned.csv")

```

Output:





Result:

EDA was successfully performed and the cleaned dataset was saved.

